

AGAI: A ResNet-Anchored, RF-DETR-Grounded Dual-Model Pipeline for Strawberry Disease Diagnosis and Report Generation

Current Technical Report and Reproducible System Status

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Abstract

This paper reports the current state of AGAI, a strawberry disease diagnosis pipeline that couples a ResNet-50 classifier, RF-DETR part grounding, and a report-generation stage built on MiniGPT-v2. The current dataset has been rebalanced in two ways. First, the active holdout now contains 20 images for each of the seven classes, giving a balanced 140-image evaluation split. Second, the augmented training split has been topped up to a minimum of 300 images for every underrepresented class using lightweight deterministic image augmentation. On this balanced holdout, the current ResNet v3 checkpoint reaches 83.57% top-1 accuracy, 92.14% top-2 accuracy, and 6.15% expected calibration error under two-view test-time augmentation and temperature scaling. Gray mold and root rot each reach 100.0% top-1 accuracy, healthy reaches 95.0%, frost and white mold each reach 85.0%, overwatered reaches 70.0%, and drought reaches 50.0%.

The runtime stack includes a threshold-based acceptance gate, RF-DETR part grounding, optional knowledge-graph retrieval, and heuristic hallucination-retry logic. On the language side, the system now has a tightened offline dual-report generation path that uses GPT-5.4 only during dataset construction. The current pilot generates one standard farmer-facing report for every image and conditionally generates a second ambiguity-aware report only for semantically realistic close pairs under a strict margin policy. The focused pilot completed successfully with 36 generated records from 20 images, including 2 tightly gated ambiguity reports and no fallback generations.

The resulting manuscript is a current technical paper for the project as it exists now. The classifier side is measured on the active balanced split. The report-generation side has a completed pilot and a clear next step for full stage-2 regeneration, retraining, and clean holdout explanation evaluation.

1 Introduction

Automated plant disease diagnosis has two separate failure modes. The first is plain classification error. The second is report failure, where a language model produces a detailed explanation that

sounds useful but is not actually supported by the image. In agriculture, the second failure mode matters as much as the first. A confident but wrong management recommendation can be more damaging than an honest uncertain answer.

The AGAI project addresses that problem by separating diagnosis from report writing. A ResNet-50 classifier provides the categorical diagnosis anchor, RF-DETR grounds the visible plant parts that the report is allowed to discuss, and MiniGPT-v2 is treated as a report writer whose job is to explain and contextualize the diagnosis rather than replace it. A disease knowledge graph can still provide supplemental text support, but it is downstream of the core diagnosis-and-grounding path. This architecture was chosen for one reason: in a safety-sensitive domain, the system should have one explicit diagnostic authority and one explicit visual-grounding stage instead of allowing a vision-language model to improvise a label or evidence trail from scratch.

That architectural choice only matters if the train and evaluation splits are honest. Much of the recent engineering effort in AGAI has therefore focused on split integrity and artifact validity. The project now uses a balanced 20-per-class holdout, an augmented training split with a minimum class floor for previously underrepresented classes, explicit provenance for holdout moves and augmentation descendants, and a current classifier checkpoint retrained on that active dataset state.

At the same time, the language-model side is in transition. The old stage-2 dataset encoded inference-heavy reports that were often too specific for what the image actually showed. The current work has therefore shifted to a stricter stage-2 generation policy: full farmer-facing diagnostic reports, tighter visual grounding, explicit separation between interpretation and visible evidence, and a heavily constrained ambiguity mode that only survives for plausible close pairs. That redesign has been piloted, but not yet scaled into a full retrain on the current balanced dataset state.

The purpose of this paper is to document the project at its current scientific state. The classifier results are clean, measured, and directly tied to the live code. The runtime architecture is deployed. The grounded report-generation redesign has completed a pilot stage and is ready for full dataset regeneration and retraining. The resulting paper is therefore strongest when it emphasizes verified results first and treats ongoing report-generation work as exactly that: ongoing work.

This paper makes four concrete contributions:

1. It documents a balanced seven-class classifier evaluation with 20 holdout images per class and an augmented training split brought to a minimum of 300 images for each underrepresented class.
2. It formalizes the current runtime diagnosis and gating stack, including calibration, thresholding, RF-DETR grounding, optional retrieval, and heuristic retry behavior.
3. It describes and measures the current offline dual-report generation policy, including the tightened ambiguity logic that only allows semantically realistic alternatives under a strict margin gate.
4. It separates completed and pending work explicitly, so the manuscript can function as a reliable current-state technical report rather than an outdated project proposal.

2 Related Work

Image-based plant disease classification has been studied extensively, especially since the availability of curated crop image datasets and transfer learning from ImageNet-scale backbones. Mohanty et al. showed that deep convolutional networks can perform strongly on plant disease datasets when

given labeled images and sufficient transfer-learning support [1]. That result helped establish the basic feasibility of smartphone-style plant diagnosis, but most work in that tradition terminates at classification. It does not address how a system should explain its answer, how it should qualify uncertainty, or how it should avoid detailed but unsupported claims.

ResNet remains a strong baseline for this setting because its residual formulation supports deep feature extraction without the optimization instability that affected earlier very deep CNNs [2]. In AGAI, the important point is not that ResNet is novel. The important point is that a calibrated ResNet-50 can serve as a high-precision disease anchor while the language model is constrained to explain rather than relabel.

Calibration matters because a diagnostic system should use probabilities in a meaningful way, not merely output them. Guo et al. showed that modern neural networks are often miscalibrated and that temperature scaling provides a simple and effective post-hoc correction [3]. AGAI follows that practical recipe. Two-view test-time augmentation and temperature scaling are used to produce the classifier probabilities that drive both the runtime acceptance gate and the offline ambiguity pilot.

On the language side, the current system inherits its report-writing backbone from the MiniGPT line of models [4, 5], which align a vision encoder with a large language model. The active repository configuration uses Llama 2 as the language backbone [6] and LoRA for parameter-efficient adaptation [7]. These choices are pragmatic rather than novel. The distinctive feature in AGAI is the surrounding policy: the language model is not treated as a free diagnostician. It is trained and prompted as a constrained report generator.

That distinction connects AGAI to broader work on guardrails and grounded generation. In mission-critical domains such as medicine, safety, and defense, a model that expresses uncertainty honestly can be more useful than a model that simply maximizes fluency. The ambiguity work reported here should be read in that tradition. The goal is not to let the language model wander through the seven disease classes. The goal is to teach it a narrow and useful uncertainty behavior when the classifier itself exposes a plausible close alternative.

AGAI still differs from many guardrail settings in one important respect. The live product must remain fully local and free at inference time. GPT-5.4 is used only offline during stage-2 dataset generation. That means any improvement has to survive deployment without access to GPT or external scoring signals. The system must therefore learn better behavior from better targets, not from a runtime API dependency.

3 Dataset Curation and Split Integrity

The current AGAI dataset is organized around seven strawberry health classes: *drought*, *frost*, *gray mold*, *healthy*, *overwatered*, *root rot*, and *white mold*. The project uses two training roots. The smaller categorical training split stores the compact base-image pool. The larger augmented split is the active classifier-training and stage-2 generation root. The active holdout is a physically separate seven-class tree.

The current counts are shown in Table 1. Figure 1 visualizes the same class counts, and Figure 2 shows representative holdout examples.

Table 1: Current dataset counts after the balanced holdout rebuild, leakage cleanup, augmentation top-up, and detector-audit imports.

Class	train	train_aug	holdout
Drought	46	300	20
Frost	94	300	20
Gray mold	592	592	20
Healthy	312	312	20
Overwatered	61	300	20
Root rot	119	300	20
White mold	206	300	20
Total	1430	2404	140

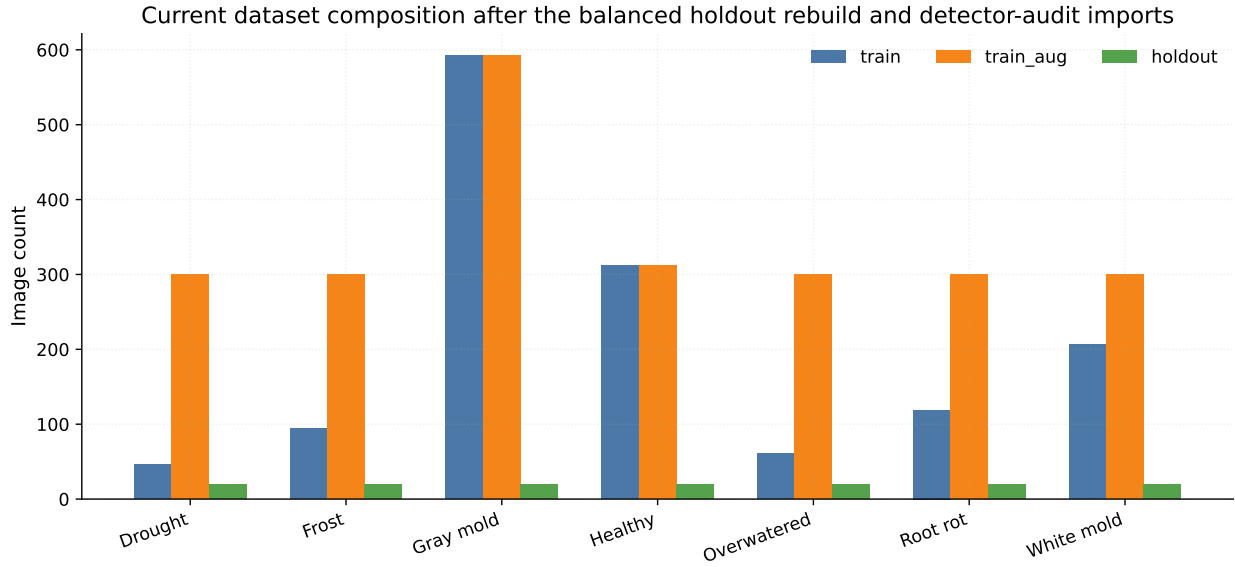


Figure 1: Current per-class image counts in the compact training split, the augmented training split, and the active balanced holdout. The augmented split now enforces a minimum of 300 images for each underrepresented class.

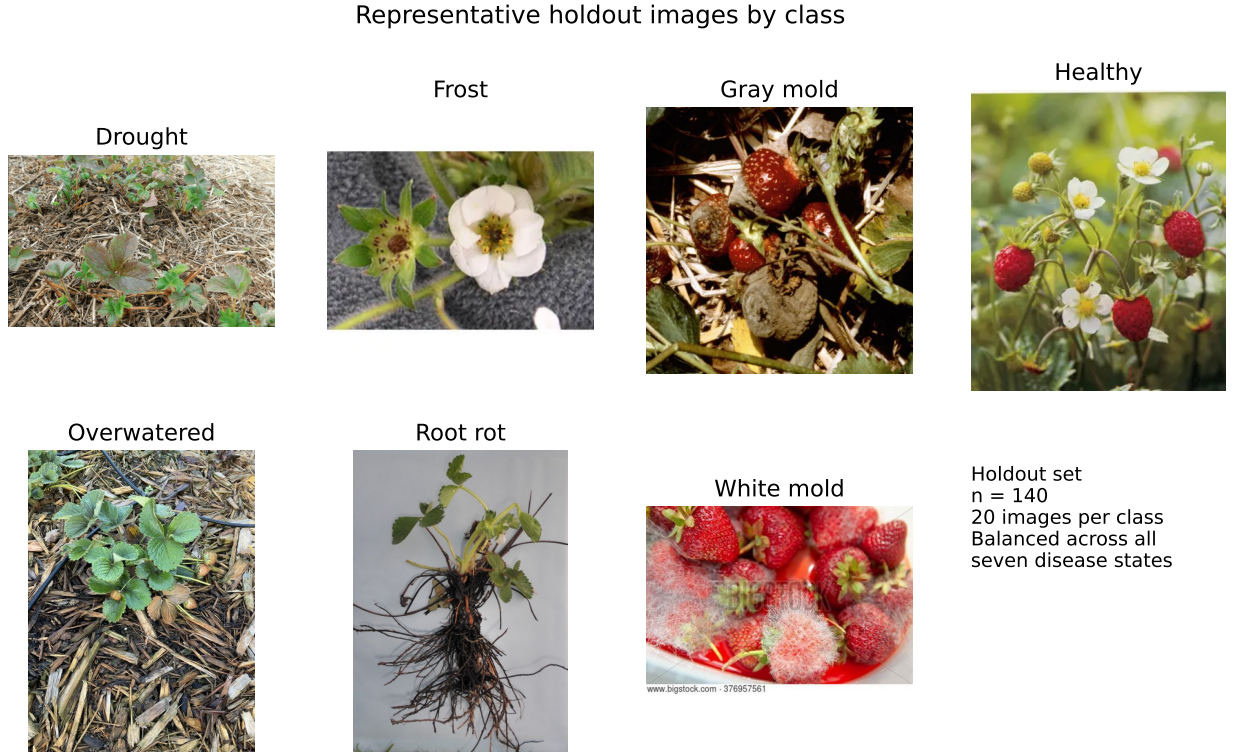


Figure 2: Representative holdout images. The active holdout contains 20 images for each of the seven classes.

3.1 Balanced Holdout Construction

The active holdout was created by physically moving source images out of the training roots rather than relying on a logical exclusion manifest. This design was chosen deliberately. A logical manifest can fail open when a future training script simply points a default image loader at a directory tree. A physical move fails closed. If the file is not in the training root, a default loader cannot silently leak it back into training.

The current holdout contains 20 images per class. For the mold classes, the repository state was especially clean because gray mold and white mold originally matched one-to-one across the compact and augmented training roots. For the non-mold classes, the resplit had to account for augmentation descendants in the augmented root. In those cases, a holdout source stem was removed together with its associated `_augNNN` descendants so that the active holdout would not remain partially represented in the training pool.

3.2 Balanced Augmented Training Split

The augmented training split was then topped up to a minimum of 300 images for each underrepresented class using lightweight deterministic image augmentation. The balancing transforms are intentionally simple: horizontal flips, small rotations, mild brightness or contrast changes, and a small flip-plus-rotation composite. This balancing step does not alter the high-support classes that already exceeded the target floor. It only raises the classes that would otherwise remain substantially underrepresented after the holdout carve.

Current AGAI pipeline. Solid arrows denote deployed runtime flow. Dashed arrows denote the offline data-generation path used for the tightened dual-report pilot. Clean post-carve MiniGPT retraining and holdout explanation evaluation are still pending.

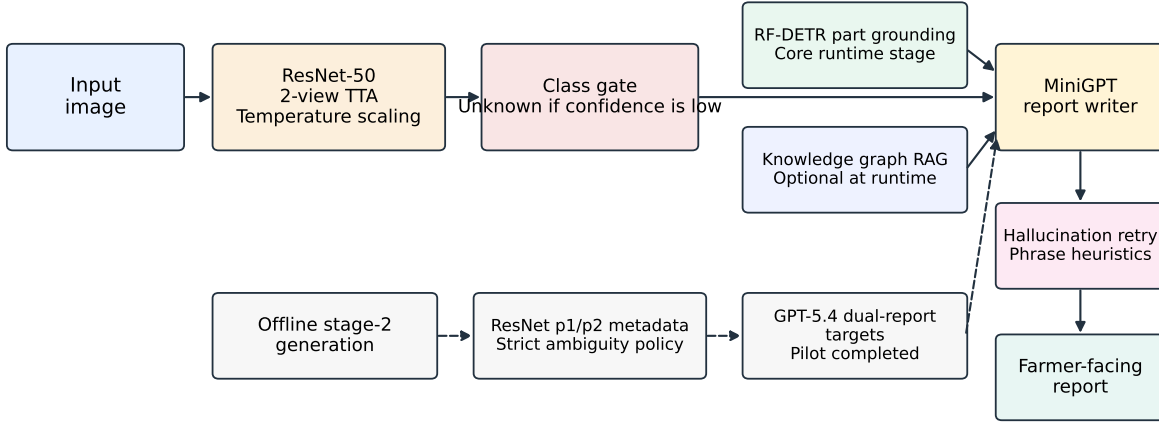


Figure 3: Current AGAI pipeline. The classifier and runtime prompt assembly are deployed. The dual-report path is currently an offline pilot for stage-2 regeneration. Clean MiniGPT retraining and clean holdout explanation evaluation are still pending.

3.3 Detector-Audit Imports

In addition to the resplit and balancing work, a separate human audit of detector-only images contributed 18 labeled images back into the training pools. Those imports were not symmetric across classes. They added 1 drought image, 5 gray mold images, 3 healthy images, 1 overwatered image, and 8 root rot images. There were no white mold additions in that batch. Those images remain part of the current training state.

3.4 What This Fixes

The current dataset state closes two different problems. First, it removes the earlier situation in which the active holdout overrepresented the mold classes while leaving very small supports for the non-mold classes. Second, it removes training leakage for holdout source stems and their augmentation descendants. The result is a balanced 140-image holdout and an augmented training split that no longer leaves frost, white mold, drought, overwatered, or root rot below the minimum class floor.

4 System Architecture and Current Methods

Figure 3 summarizes the current AGAI pipeline. Solid arrows represent the deployed runtime path. Dashed arrows represent the offline stage-2 data-generation path used for the current dual-report pilot.

4.1 Classifier Backbone, Test-Time Augmentation, and Calibration

The active classifier is a ResNet-50 [2]. The repository training script uses standard ImageNet normalization, moderate geometric and photometric augmentation during training, and a 256-pixel evaluation crop. At inference time, the deployed classifier averages two views: the original center-cropped image and its horizontal flip. If the raw logit vector for image x is $z(x) \in \mathbb{R}^7$ and the learned temperature is $T > 0$, then the deployed class distribution is

$$\bar{p}(c \mid x) = \frac{1}{2} \left[\text{softmax} \left(\frac{z(x)}{T} \right)_c + \text{softmax} \left(\frac{z(\text{flip}(x))}{T} \right)_c \right]. \quad (1)$$

The predicted class is the maximum-probability class under $\bar{p}(c \mid x)$:

$$\hat{y}(x) = \arg \max_{c \in \mathcal{C}} \bar{p}(c \mid x), \quad (2)$$

where $\mathcal{C}$ is the seven-class label set.

The current balanced-split checkpoint uses a learned temperature of 0.7810 on the active holdout evaluation. The deployed classifier continues to use the same two-view TTA policy.

4.2 Runtime Acceptance Gate

The runtime application does not accept every classifier output as an actionable diagnosis. It applies a per-class threshold gate:

$$\text{accept}(x) = \mathbf{1} \left[\bar{p}(\hat{y}(x) \mid x) \geq \tau_{\hat{y}(x)} \right]. \quad (3)$$

The current thresholds are

$$\begin{aligned} \tau_{\text{healthy}} &= 0.40, & \tau_{\text{overwatered}} &= 0.60, & \tau_{\text{root rot}} &= 0.65, \\ \tau_{\text{drought}} &= 0.65, & \tau_{\text{frost}} &= 0.70, & \tau_{\text{gray mold}} &= 0.60, & \tau_{\text{white mold}} &= 0.60. \end{aligned} \quad (4)$$

The default threshold for any unmapped class is 0.55. A second conservative guard in the runtime code demotes the prediction to **unknown** when the accepted top-1 probability falls below 0.50 after prompt assembly.

The system also logs p_1 , p_2 , and the margin $p_1 - p_2$ for later analysis. A margin threshold of 0.08 exists in the codebase only as a logging quantity. It is not currently applied at runtime. This is important because the current classifier claims in this paper are threshold-only runtime claims, not threshold-plus-margin claims.

4.3 Grounded Runtime Report Path

The report-generation stage is deliberately downstream of the classifier. The ResNet label is injected into the prompt as the authoritative diagnosis, RF-DETR contributes the visible-part and not-visible-part evidence summary, and the report writer is expected to explain that grounded diagnosis rather than re-predict it.

The runtime stack has one required grounding stage and two additional support mechanisms:

1. **RF-DETR part grounding.** The detector is part of the runtime pipeline. Its part detections are converted into visible-part and not-visible-part lists that constrain what the report is allowed to claim about the image.
2. **Knowledge-graph retrieval.** If the local disease QA resources load successfully, retrieved snippets can be appended to the report prompt as supplemental context.
3. **Heuristic hallucination retry.** A phrase-triggered retry gate attempts to catch certain known overclaims by matching hard-coded substrings such as disease names or visual claims. This is currently heuristic and not yet calibrated against a clean false-positive budget.

All three components operate locally at runtime. GPT is not used at inference time.

4.4 Current MiniGPT Configuration

The current MiniGPT training configuration uses a MiniGPT-v2 architecture [5] with a Llama 2 chat backbone [6] and LoRA adaptation [7]. The active settings specify a 448-pixel image size, maximum text length 500, LoRA rank 64, LoRA alpha 128, dropout 0.05, batch size 2 per GPU, 8 gradient accumulation steps, and 20 epochs with a 10% auto-validation split. These are the active training settings in the repository, not proof that a clean MiniGPT retrain on the newly balanced stage-2 dataset has already been completed.

4.5 Offline Dual-Report Generation Policy

The current stage-2 redesign uses GPT-5.4 only offline. Every image receives one standard farmer-facing report. Some images also receive a second report, but only under a deliberately strict ambiguity policy.

Let the ground-truth folder label be $g(x)$, the top-1 classifier label be $y_1(x)$, and the top-2 classifier label be $y_2(x)$. The strongest competing class used for the offline policy is

$$a(x) = \begin{cases} y_2(x), & y_1(x) = g(x), \\ y_1(x), & y_1(x) \neq g(x). \end{cases} \quad (5)$$

The second report is allowed to be ambiguity-aware only when the candidate pair is semantically plausible and the probabilities are genuinely close. The current allowed pair set is

$$\mathcal{A} = \left\{ \begin{array}{l} \{\text{gray mold, white mold}\}, \\ \{\text{drought, overwatered}\}, \\ \{\text{drought, frost}\}, \\ \{\text{healthy, overwatered}\}, \\ \{\text{root rot, overwatered}\} \end{array} \right\}. \quad (6)$$

Under the current builder, a second report is marked *ambiguous* only if one of two cases holds:

$$\begin{aligned} y_1(x) = g(x), \quad \bar{p}(y_1 | x) - \bar{p}(y_2 | x) &\leq 0.10, \\ \bar{p}(y_2 | x) &\geq 0.15, \quad \{g(x), y_2(x)\} \in \mathcal{A}, \end{aligned} \tag{7}$$

$$\begin{aligned} y_1(x) \neq g(x), \quad \text{rank}(g(x)) &= 2, \\ \bar{p}(y_1 | x) - \bar{p}(g | x) &\leq 0.10, \\ \bar{p}(g | x) &\geq 0.15, \quad \{g(x), y_1(x)\} \in \mathcal{A}. \end{aligned} \tag{8}$$

If the classifier is correct and the margin is clear, the second report becomes a *standard variant* rather than an ambiguity report. If the classifier is wrong and the ground truth is not a close rank-2 competitor, the second report is skipped entirely. This prevents obvious classifier failures from being laundered into fake ambiguity.

This policy is currently a pilot-stage data-generation policy only. It is not yet a claim about the live runtime behavior of the deployed report generator.

5 Current Experimental Results

The current results divide naturally into two layers. The classifier layer is fully measured on a clean seven-class holdout. The report-generation layer has completed a tightened pilot for stage-2 target construction, but its full regenerated training set, retrained checkpoint, and clean holdout explanation metrics are still in progress. The sections below therefore emphasize completed classifier results first and then treat the dual-report pilot as a dataset-generation result rather than a finished end-to-end benchmark.

5.1 Clean ResNet Holdout Results

The current classifier summary is given in Table 2. The active ResNet v3 checkpoint is evaluated on the balanced 140-image holdout with 20 images per class.

Table 2: ResNet holdout summary for the current balanced-split checkpoint.

Metric	Current v3
Holdout samples	140
Top-1 accuracy	83.57%
Top-2 accuracy	92.14%
ECE	6.15%
Temperature	0.7810
Horizontal-flip TTA	yes

The most important property of this result is not that it is numerically higher than a previous split. It is that it is measured on the current active dataset state: balanced holdout support, balanced augmented training counts for underrepresented classes, and explicit exclusion of holdout source stems from the training roots. On this balanced holdout, the current checkpoint reaches 83.57% top-1 accuracy, 92.14% top-2 accuracy, and 6.15% expected calibration error.

Table 3: Per-class holdout performance for the current balanced-split ResNet checkpoint.

Class	n	Top-1	Wilson 95% CI	Top-2
Drought	20	50.0%	[29.9, 70.1]	80.0%
Frost	20	85.0%	[64.0, 94.8]	85.0%
Gray mold	20	100.0%	[83.9, 100.0]	100.0%
Healthy	20	95.0%	[76.4, 99.1]	100.0%
Overwatered	20	70.0%	[48.1, 85.5]	85.0%
Root rot	20	100.0%	[83.9, 100.0]	100.0%
White mold	20	85.0%	[64.0, 94.8]	95.0%

Per-class metrics for the current checkpoint are given in Table 3. Because the holdout now allocates 20 images per class, the class comparisons are much easier to interpret than they were under the earlier mold-heavy split. Gray mold and root rot each reach 100.0% top-1 accuracy on 20 images, healthy reaches 95.0%, frost and white mold each reach 85.0%, overwatered reaches 70.0%, and drought reaches 50.0%.

Figure 4 shows the current confusion matrix. Figure 5 shows the per-class top-1 accuracies and the current reliability diagram.

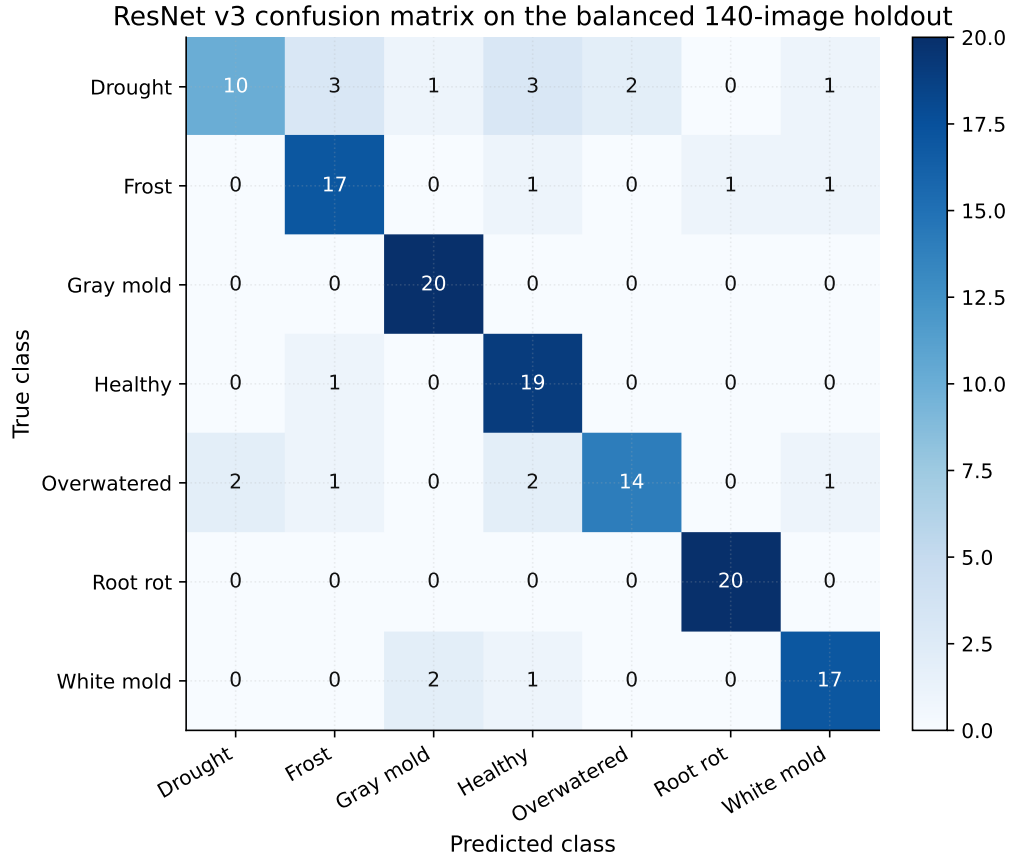


Figure 4: Confusion matrix for the current ResNet checkpoint on the balanced 140-image holdout. The balanced split makes the weaker drought and overwatered rows immediately visible.

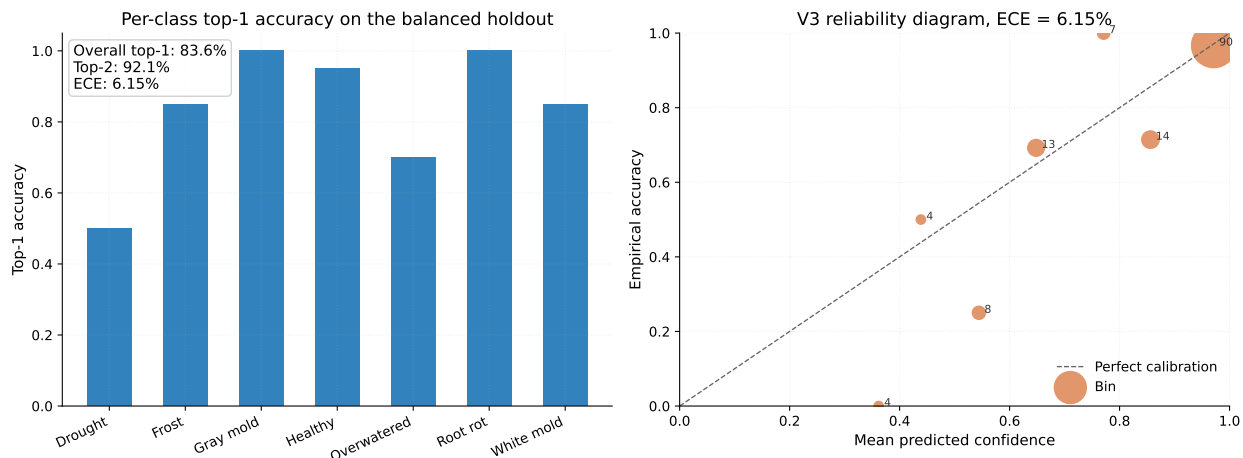


Figure 5: Left: per-class top-1 accuracy on the current balanced holdout. Right: the current reliability diagram. The current checkpoint is better calibrated than the earlier mold-heavy split under expected calibration error, even though the balanced holdout is harder overall.

5.2 Runtime Gating Behavior and Top-2 Structure

The classifier accuracy story is not the whole runtime story. A report can still be blocked if the threshold gate demotes a correct label to **unknown**. That gate remains threshold-only in the live runtime, with p1/p2 margins recorded for later analysis but not yet activated as a runtime demotion rule. The current paper therefore treats the holdout metrics as classifier metrics first, not as final deployed diagnosis rates.

The top-2 structure remains informative because the ambiguity pilot is built around close alternatives rather than free-form speculation. The report-generation policy still only allows ambiguity for semantically realistic disease pairs under a strict margin rule, and the current pilot should be read as a controlled dataset-generation experiment rather than a user-facing ambiguity feature.

5.3 Tight Dual-Report Pilot

The current dual-report pilot is not a full dataset regeneration. It is a focused evaluation of whether the stage-2 policy produces useful training targets rather than boilerplate or semantically absurd alternatives.

The tight pilot summary is given in Table 4 and Figure 6. The pilot selected 20 images and generated 36 total records: 20 standard reports, 14 standard variants, and 2 ambiguity reports. No fallback generations were required.

Table 4: Summary of the current focused tight dual-report pilot.

Focused tight dual-report pilot statistic	Value
Selected images	20
Generated records	36
Hard-case selections	4
Close-margin selections	2
Standard reports	20
Standard variants	14
Ambiguous reports	2
Self-reported low overclaim risk	28
Self-reported medium overclaim risk	8
Fallback reports	0

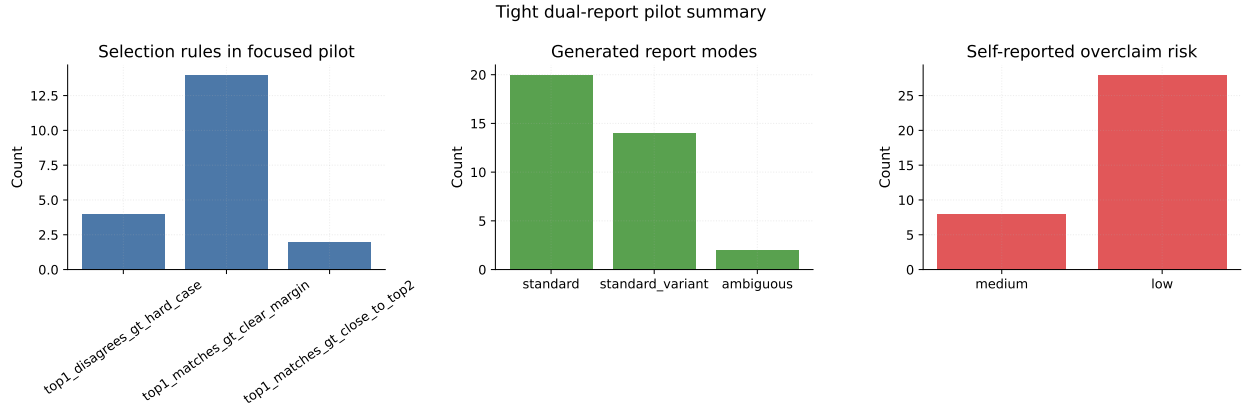


Figure 6: Summary of the current tight dual-report pilot. After the ambiguity policy was tightened, ambiguity reports became rare and were restricted to semantically plausible close pairs.

The important qualitative outcome is that the tightened policy eliminates bizarre alternatives that arise from raw numeric proximity alone. Under the current strict margin rule and pair whitelist, the surviving ambiguity cases in the focused pilot are disease pairs that are visually and operationally plausible, rather than arbitrary second-ranked classes.

This pilot therefore supports three limited claims:

1. The dual-report plumbing works end to end.
2. The strict ambiguity policy can suppress obviously implausible alternatives.
3. The current pilot outputs are good enough to motivate full stage-2 regeneration.

It does *not* support a clean claim about final explanation quality on the post-carve holdout, because the current deployed MiniGPT checkpoint predates the full regeneration and retrain that would be required for that claim.

5.4 What Is Still Not Measured

Several experiments remain deliberately unreported because they are not yet clean.

First, the current stage-2 training annotations do not yet encode the tightened dual-report policy at full scale. Second, the current MiniGPT checkpoint predates the balanced split and the current stage-2 redesign. Third, no clean holdout explanation rubric has yet been applied to a retrained post-regeneration checkpoint. Fourth, the RF-DETR on/off ablation is planned but not yet executed on that clean holdout. Those missing steps are not minor bookkeeping. They are the difference between a promising report-generation pilot and a finished end-to-end explanation result.

6 Discussion

The current system supports a stronger classifier claim than language-model claim, and the paper should say so plainly.

On the classifier side, the methodology is now substantially cleaner than it was during the earlier development phase. The active holdout is physically separate, the current split support is balanced across classes, the augmented training split has been raised to a minimum floor for the previously underrepresented classes, and the current checkpoint is evaluated on that active dataset state. That is enough to support the classifier results in this manuscript with confidence.

On the language side, the story is more nuanced. The runtime integration of MiniGPT, RF-DETR, and RAG is real. The prompt-design work is real. The dual-report pilot is real. But the current MiniGPT checkpoint still predates the fully regenerated balanced stage-2 dataset, which means any strong claim about holdout explanation quality would still be contaminated by outdated supervision. The correct interpretation is therefore that the report-generation redesign is promising but not yet complete enough for a final benchmark claim.

The current threshold policy still deserves separate treatment from raw classifier accuracy. The classifier can be correct while the deployment gate suppresses a diagnosis, and the converse can also happen on harder cases. This is a reminder that classifier quality and deployed diagnosis rate are not identical. The current paper does not retune those thresholds because the project now has p1/p2 runtime logging and should use real decision data rather than intuition for future retuning.

The ambiguity work is valuable for the same reason. A naive “top-2 means possible alternative” rule is not safe. Raw ResNet probabilities can surface numerically close but semantically implausible competitors. The pilot only became credible after the policy was tightened with both a lower margin threshold and a realistic pair whitelist. That is exactly the sort of upstream-downstream interaction that matters in a system paper: the usefulness of ambiguity is not determined by the language model alone. It depends on how the classifier’s uncertainty structure is filtered before it ever becomes text.

Finally, this project should be judged against the right target. The goal is not to maximize fluent prose. The goal is to produce a usable agricultural diagnostic system in which the diagnosis is anchored, the report is grounded, and uncertainty is surfaced honestly. Under that definition, the current paper is strongest when it reports the measured classifier and pilot results cleanly, while reserving full end-to-end language claims for the retrained model and the clean holdout evaluation that still need to be completed.

7 Conclusion

AGAI now has a clean and defensible classifier core. The current ResNet v3 checkpoint is trained on the balanced augmented split and reaches 83.57% top-1 accuracy and 92.14% top-2 accuracy on the balanced seven-class holdout, with especially strong performance on gray mold, root rot, and healthy examples. The runtime system already supports classifier-anchored reporting, RF-DETR grounding, optional knowledge-graph retrieval, and classifier decision logging.

At the same time, the report-generation side is in the middle of an important transition. The project has moved beyond free-form paragraph targets toward a stricter farmer-facing report format and a tightly constrained ambiguity policy. The focused dual-report pilot indicates that this direction is useful and that ambiguity can be kept rare, believable, and semantically grounded. However, the full stage-2 regeneration on the current balanced split, the corresponding MiniGPT retrain, and the clean holdout explanation evaluation are still pending.

The paper therefore closes on a simple factual statement. AGAI is already a real system with a measured and methodologically clean classifier component. Its grounded reporting redesign is promising and partially implemented, but not yet finished enough to support a clean end-to-end explanation claim. That is not a weakness of the manuscript. It is the reason the manuscript is trustworthy.

A Appendix: Configuration and Project Scope Notes

A.1 Completed Versus Ongoing Work

Table 5 summarizes the current project state. The classifier side is complete enough to support the results reported in the main text. The report-generation side has completed a pilot but still needs full stage-2 regeneration, clean retraining, and clean holdout evaluation before it can support a final end-to-end claim.

Table 5: Current project status summary.

Component	Status	Notes
Balanced holdout rebuild with provenance	Done	The active holdout now contains 20 images for each of the seven classes
Augmented class-floor balancing	Done	Underrepresented classes in the active augmented split were raised to a minimum of 300 images
ResNet v3 retrain on the balanced split	Done	The current classifier checkpoint is evaluated on the balanced 140-image holdout
Runtime promotion with rollback path	Done	The runtime app defaults to the current v3 checkpoint and keeps an explicit override path
p1/p2 runtime logging	Done	JSONL logging added for downstream calibration analysis
Tight dual-report generation pilot	Done	Focused 20-image pilot completed with 36 generated reports
Full stage-2 regeneration under tight policy	Pending	The current stage-2 JSON remains stale relative to the augmented training root
MiniGPT retrain on regenerated stage-2	Pending	Current checkpoint predates the holdout carve and dual-report policy
Clean holdout explanation evaluation	Pending	No post-regeneration holdout explanation metric exists yet
RF-DETR on/off ablation on clean holdout	Pending	Planned but not executed

A.2 MiniGPT Training Configuration in the Repository

The active MiniGPT training configuration in the repository uses:

- image size 448 for both train and evaluation processors,
- maximum text length 500,
- LoRA rank 64, LoRA alpha 128, and dropout 0.05,
- initial learning rate 10^{-5} ,
- 20 epochs,
- per-GPU batch size 2 with 8 gradient accumulation steps,
- a 10% auto-validation split.

These are active repository settings, not proof of a clean MiniGPT training run on the current balanced stage-2 dataset. The current configuration still points to the older stage-2 annotation path, which is why the clean regeneration remains a pending item in Table 5.

A.3 Runtime Thresholds

The current runtime thresholds are listed in Equation 4. The live application currently uses threshold-only acceptance, logs p1/p2 margins for later analysis, and applies a second conservative low-confidence demotion when the accepted p1 value falls below 0.50. The margin rule is intentionally not active yet because its operating point has not been calibrated on sufficient real decision logs.

A.4 Ambiguity Policy Scope

The current ambiguity pilot exists only in the offline dataset-generation path. It does not yet imply that the runtime report generator should present possible alternatives directly to farmers. Before that can become a product claim, the project still needs a clean post-regeneration MiniGPT retrain and a clean holdout explanation evaluation with and without RF-DETR grounding.

References

- [1] Sharada Prasanna Mohanty, David Hughes, and Marcel Salathé. Using deep learning for image-based plant disease detection, 2016. URL <https://arxiv.org/abs/1604.03169>.
- [2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016. URL <https://arxiv.org/abs/1512.03385>.
- [3] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.
- [4] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigpt-4: Enhancing vision-language understanding with advanced large language models, 2023. URL <https://arxiv.org/abs/2304.10592>.
- [5] Jun Chen, Deyao Zhu, Xiaoqian Shen, Xiang Li, Zechun Liu, Pengchuan Zhang, Raghuraman Krishnamoorthi, Vikas Chandra, Yunsang Xiong, and Mohamed Elhoseiny. Minigpt-v2: large language model as a unified interface for vision-language multi-task learning, 2023. URL <https://arxiv.org/abs/2310.09478>.
- [6] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Danila Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Vikramjit Khandelwal, Manohar Paluri, and et al. Llama 2: Open foundation and fine-tuned chat models, 2023. URL <https://arxiv.org/abs/2307.09288>.
- [7] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL <https://arxiv.org/abs/2106.09685>.